Quiroga–FRAMES best-for-best setup

Campaign setup `qpt_quiroga_best_for_best_20261004_02` · 4 October 2026

LOCAL SETUP COMPLETE — REMOTE SMOKE NOT SUBMITTED.

Pinned source revision: `0586c15a8c4eccf644b67d3bae97dc3b017e6e69`.

This bundle corrects the primary comparison so that each method receives its own calibrated TP treatment. Fairness comes from shared physical data and evaluation—not from forcing two different algorithms to optimize the same penalized objective.

Validation completed

Focused tests	27 passed, 0 failed
Full repository suite	532 passed, 55 skipped, 0 failed
Independent audit	Approved
Committed source	`0586c15a8c4eccf644b67d3bae97dc3b017e6e69`

Method-native objectives

Method	Objective and independently selected controls
Printed adaFGD	$0.5\|\|r\|\|^2 + \lambda_Q H$, with fixed $\alpha_Q=\lambda_Q/m$. Select α_Q and `eta_scale` independently.
FRAMES	$(1/(2m))\|\|r\|\|^2 + \alpha_{F,k} H$, with $\alpha_{F,0}=1/(2\beta_0)$ and $\alpha_{F,k}=\alpha_{F,0}(k+1)^{0.25}$. Select only the initial $\alpha_{F,0}$ from its own grid; the exponent is fixed.

Critical analysis rule: method-native penalized objectives are retained as diagnostics

but are never differenced, pooled, or used as a cross-method score. Paired differences are limited to recovery/physical metrics: loss or RMSE, fidelity, TP violation, and the gate indicator. Factor norms, timing, and work counters remain method diagnostics—not paired estimands.

Predeclared grids

Printed α_Q	0.5, 0.05, 0.005, 0.0005, 0.00005
Printed `eta_scale`	1e-4, 3e-4, 1e-3, 3e-3, 1e-2, 3e-2
FRAMES $\alpha_{F,0}$	0.5, 0.05, 0.005, 0.0005, 0.00005
FRAMES exponent	`p=0.25`, fixed and positive

The two alpha grids happen to contain the same numeric values, but they are independent method-specific grids and do not define paired penalty settings.

Shared comparison contract

- Identical data/channel instance, active rows, observations, serialized `U0`, rank, precision, logical row budget, and checkpoints.
- Common noiseless recovery gate: fidelity ≥ 0.99, full-design RMSE ≤ 1e-6, and normalized TP violation ≤ 1e-3.
- FRAMES retains its operator-norm-ball domain; printed adaFGD remains unconstrained. This domain difference must be reported.
- Logical row epochs are the progress axis, not a compute-equivalence claim. Wall time, LMO work, and spectral-norm work remain separate.

Order and freezing

1. Run **n=2 full 128 rows**, then **n=2 subset 96**. The subset gate uses RMSE on all 128 rows.
2. Select one configuration per method on calibration instances using the predeclared gate-first rule.

3. Write and hash an immutable selection manifest containing both selected config IDs. Validation runs use those frozen IDs and cannot rerank them.
4. Run held-out instances only with the same selection-manifest hash, selected config IDs, budgets, checkpoints, gate, and source commit.
5. Proceed to **n=4 full 8,192 rows** only after the n=2 launcher, archive, accounting, and aggregation checks pass.

Previous corrected-design runs remain preserved as exploratory or ablation evidence. The old shared-penalty bundle is preserved unchanged but is superseded for primary claims; it may be reported only as a matched-penalty ablation. Legacy high-qubit QPT_BFW runs remain valid only for their local-Pauli sensing design.

Local n=2 full-design smoke

Outcome	Completed; both methods complete; schema 2; protocol `method_tuned_dynamic_frames_matched_nominal_row_accesses_v2`
Printed adaFGD	`alpha_Q=0.05`; fixed `lambda_Q=6.4`
FRAMES initial	`alpha_F0=0.005`; `beta0=100`; `p=0.25`
FRAMES endpoint	`alpha_F=0.008408964152537144`; `beta=59.46035575013606`
JSON SHA-256	`3f68231ff64c0cff1bb7b725c30940d20f8b4a58ea121bc863861e30fc888301`
NPZ SHA-256	`5f1f593cb27b8439fc48990bccad56ff06b3e271284c46a85d614813c8de40bc`

This two-row-epoch run validates launcher, provenance, budget, serialization, and method plumbing. It is not a recovery result.

Ruche smoke staged locally

Partition/	`cpu_short`; 1 task; 2 CPUs; 1 GiB; 20 minutes

resources	
Environment	`anaconda3/2023.09-0/none-none`, Conda environment `qpt`
Output root	`/gpfs/workdir/silvetian/qpt_quiroga_best_for_best_0586c15a8c4eccf644b67d3bae97dc3b017e6e69_20261004_02`
CLI penalty flags	`--printed-tp-weight-per-row`, `--frames-initial-tp-weight-per-row`, `--frames-penalty-exponent 0.25`
Submission state	**Not transferred or submitted.** Transfer and submission require separate authorization and remote preflight checks.

Bundle and deferred work

This HTML, its PDF rendering, the plan, launcher, and local smoke pair are sealed by `manifest.sha256`. Selection manifests, calibration/validation/held-out results, remote logs, transfer, and submission remain deliberately unperformed.